

Christoffer Björkskog · christoffer.bjorkskog@novia.fi

Lamin Jatta · lamin.jatta@novia.fi

Maritime Technology, Novia University of Applied Sciences

Supervisors: Andreas Lundell (AAU), Johan Westö (Novia UAS), Mikael Manngård (Novia UAS).

PROBLEM

Engineers still copy values from datasheets, product leaflets, P&ID drawings, and control-system manuals into spreadsheets and simulation models by hand. Normal chat answers hide where numbers came from. One wrong pressure, temperature, or dimension can travel into a calculation with no route back to the source page

WHAT ANCHOR DOES

Anchor makes document evidence visible and reusable:

- Drop technical PDFs into a shared engineering workspace.
- Convert pages into text, images, tables, figures, and semantic regions.
- Ask an agent for specs, operating limits, tables, or facts.
- Place grounded cards on the canvas with source links.
- Jump from a canvas answer back to the exact PDF page and region.
- Keep document evidence separate from FMU wiring so engineers stay in control.

HOW IT WORKS

- Bronze → Silver → Gold → Canvas
- Bronze: Raw uploaded PDFs.
- Silver: Page markdown, page images, outlines, tables, figures, and document index.
- Gold: Semantic regions such as tables, charts, diagrams, captions, text blocks, and spec blocks. Each region can carry page and bbox provenance.
- Canvas: React Flow workspace with document cards, facts, spec tables, images, FMUs, plots, notes, and visible source/evidence edges.

WHAT V0.1 SHIPS

- Upload PDFs and run the medallion document pipeline.
- Inspect bronze, silver, and gold status in the right rail.
- Ask grounded questions over documents on the canvas.
- Create fact cards, image crops, and spec/table nodes from document evidence.
- Keep source chips on answers and table rows.
- Click sources to reopen the PDF page with highlights.
- Store conversations and canvas evidence for later review.

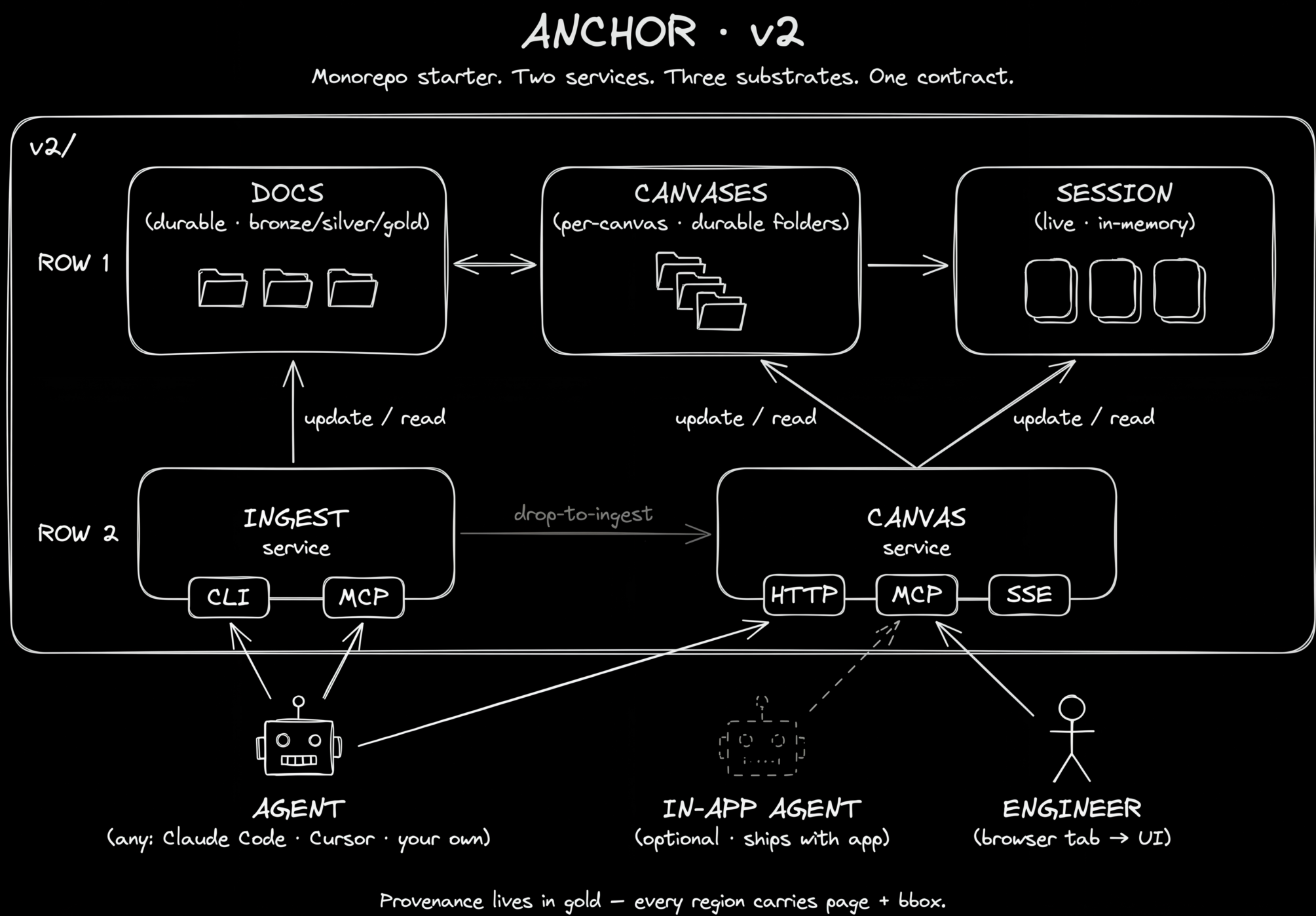
BOUNDARY

FMU upload, inspect, and simulation routes exist, and FMU nodes are part of the canvas. Automatic document-to-FMU wiring is still manual and experimental; engineers review values before wiring them into simulations.

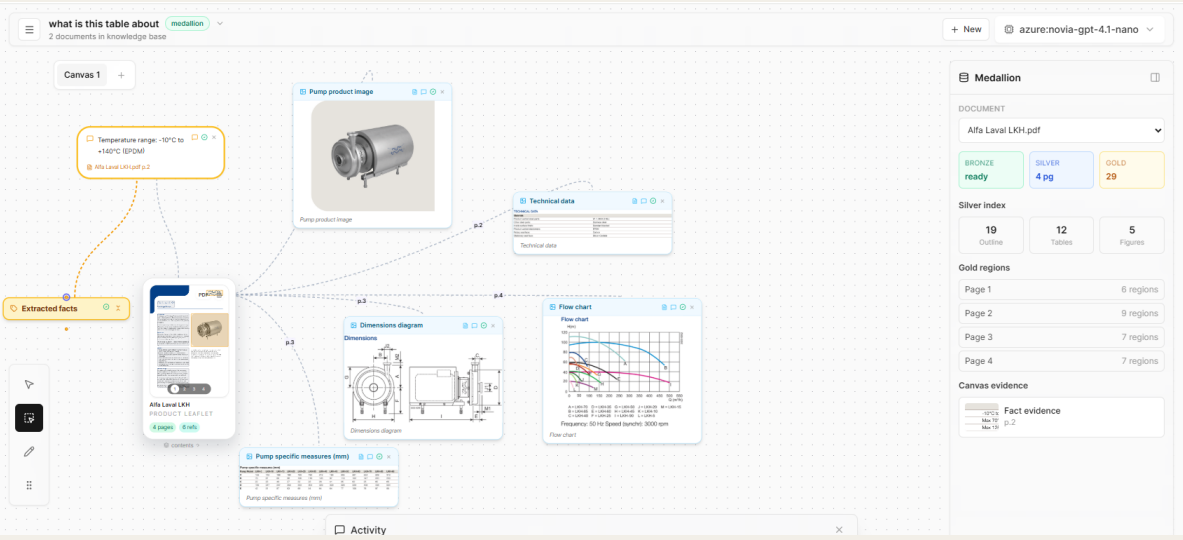
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Every spec your agent quotes points back to the page it came from.

Datasheet → canvas → simulation.
Provenance built in.



THE CANVAS



The canvas shows the proof loop: a grounded answer, its document source, PDF-derived visual regions, and the medallion status of the active datasheet.

EVIDENCE LOOP

- Ask a question about a product datasheet.
- The agent reads canvas context and document indexes.
- The answer becomes a canvas card or table.
- The source chip opens the PDF page and bbox behind the claim.

WHY INDUSTRY SHOULD CARE

- Fewer copy-paste errors from datasheets into engineering work.
- Faster review because source pages are one click away.
- Better handoff between engineers, agents, and simulation workflows.
- Local-first project data that can be inspected, copied, and versioned.
- Works with practical product leaflets, not only clean academic PDFs.

LOCAL DEMO

- Runs locally with Node.js, Python 3.12, uv, and model credentials.
- Current demo uses file-backed documents and conversations; PostgreSQL is not required.
- `git clone → npm install → uv sync → npm run dev`.

READ MORE



github.com/Novia-RDI-Seafaring/
anchor-kb-ui-RAG

Source code and local setup instructions.

Industry presentation · Vaasa, 2026